
TRIGONOMETRY

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1. IDENTITIES AND MORE

1.1 Proving Identities

A. Proving Identities

There are three main ways you can work with and prove an identity

- Work with the LHS, and show that it is equal to the RHS
- Work with the RHS and show that it is equal to the LHS

Work with both sides and show that they are equal

Example 1.1

Prove that $\tan x \csc x \cos x = 1$

$$LHS = \tan x \csc x \cos x = \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} \cdot \cos x = 1 = RHS$$

Difference of Squares:

$$(a + b)(a - b) = a^2 - b^2$$

$$(x + 4)(x - 4) =$$

Example 1.2

$$(\sin x - \cos x)(\sin x + \cos x) = 1 - 2 \cos^2 x$$

$$LHS = \sin^2 x - \cos^2 x$$

Add and subtract $\cos^2 x$:

$$= \sin^2 x + \cos^2 x - \cos^2 x - \cos^2 x$$

Use the Pythagorean Identity:

$$= 1 - 2 \cos^2 x = RHS$$

Example 1.3

$$\frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$$

Multiply the LHS with a *form of 1*:

$$LHS = \frac{\sin x}{1 + \cos x} \cdot \frac{1 - \cos x}{1 - \cos x} = \frac{\sin x (1 - \cos x)}{1 - \cos^2 x} = \frac{\sin x (1 - \cos x)}{\sin^2 x} = \frac{1 - \cos x}{\sin x} = RHS$$

Example 1.4

$$\frac{\sec x - \cos x}{\sec x} = \sin^2 x$$

$$LHS = \frac{\frac{1}{\cos x} - \cos x}{\frac{1}{\cos x}} = \left(\frac{1}{\cos x} - \cos x \right) \cos x = 1 - \cos^2 x = \sin^2 x$$

1.2 The Pythagorean Identity

A. Basics

1.5: Square of a Trigonometric Function

$$(\sin \theta)^2 = \sin^2 \theta$$

1.6: Pythagorean Identity

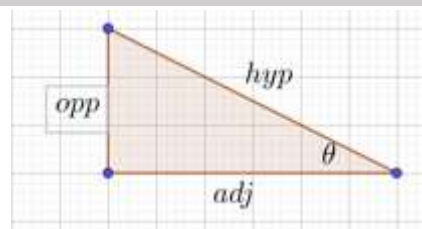
The Pythagorean identity is an important relation between $\sin \theta$ and $\cos \theta$ that is obtained using the Pythagorean Theorem:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$LHS = \sin^2 \theta + \cos^2 \theta = \left(\frac{opp}{hyp}\right)^2 + \left(\frac{adj}{hyp}\right)^2 = \frac{opp^2 + adj^2}{hyp^2}$$

By the Pythagorean Theorem, $opp^2 + adj^2 = hyp^2$:

$$= \frac{hyp^2}{hyp^2} = 1 = RHS$$



Alternate Method

Consider the unit circle.

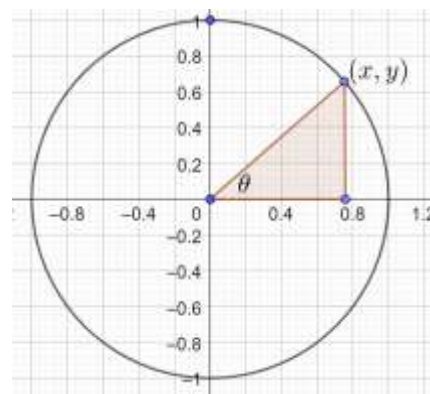
Since it is a unit circle, the length of radius is 1.

Consider any point on the unit circle, and draw a right triangle, as shown.

$$x = \cos \theta, y = \sin \theta$$

By the Pythagorean Theorem,

$$\sin^2 \theta + \cos^2 \theta = y^2 + x^2 = Hyp^2 = 1^2 = 1$$



Example 1.7

If $\cos \theta = -\frac{3}{5}$, then find $\sin \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta + \left(-\frac{3}{5}\right)^2 = 1$$

$$\sin^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$$

$$\sin \theta = \pm \sqrt{\frac{16}{25}} = \pm \frac{4}{5}$$

$\cos \theta$ is negative in 2nd Quadrant and 3rd Quadrant.

$\sin \theta$ is positive in 2nd Quadrant, and negative in 3rd Quadrant

B. Expressions

Example 1.8: Simplification

Simplify

- A. $\sin^2 x + 2 \cos^2 x$
 B. $\frac{\cot x}{\tan x}$
 C. $\sin^2 x \cos^2 x + \sin^4 x$

Part A

$$\sin^2 x + 2 \cos^2 x = \sin^2 x + \cos^2 x + \cos^2 x = \cos^2 x + 1$$

Part B

$$\frac{\cot x}{\tan x} = \cot x \times \frac{1}{\tan x} = \cot x \times \cot x = \cot^2 x$$

Part C

Factor out $\sin^2 x$ from each term:

$$\sin^2 x \underbrace{(\sin^2 x + \cos^2 x)}_{=1} = \sin^2 x$$

Example 1.9

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

Do as directed in the questions below. You might find the identities above useful.

Multiply

- A. $(\sin \phi + \cos \phi)^2$
 B. $(3 \sin \phi - 2 \cos \phi)^2$

Simplify

- C. $\cos^2 \alpha + 6 \cos \alpha + 9$
 D. $3 \tan^2 \gamma - 6 \tan \gamma + 3$
 E. $\frac{\sin^2 x + 2 \sin x \cos x + \cos^2 x}{\sin x + \cos x}$
 F. $\frac{\sin^2 x - 2 \sin x \cos x + \cos^2 x}{\sin x - \cos x}$
 G. $\frac{1 + 2 \sin \alpha \cos \alpha}{\sin \alpha + \cos \alpha}$
 H. $\frac{1 - 2 \sin \alpha \cos \alpha}{\sin \alpha - \cos \alpha}$

Part A

$$(\sin \phi + \cos \phi)^2 = \sin^2 \phi + \cos^2 \phi + 2 \sin \phi \cos \phi = 1 + 2 \sin \phi \cos \phi$$

Part B

$$\begin{aligned} & 9 \sin^2 \phi + 4 \cos^2 \phi - 12 \sin \phi \cos \phi \\ &= 5 \sin^2 \phi + 4 \sin^2 \phi + 4 \cos^2 \phi - 12 \sin \phi \cos \phi \\ &= 5 \sin^2 \phi + 4 - 12 \sin \phi \cos \phi \\ &= 5 \sin^2 \phi - 12 \sin \phi \cos \phi + 4 \end{aligned}$$

Part C

Use a change of variable. Let $a = \cos \alpha$:

$$a^2 + 6a + 9 = (a + 3)^2$$

Change back to the original variable:

$$(\cos \alpha + 3)^2$$

Part D

$$3(\tan^2 \gamma - 2 \tan \gamma + 1) = 3(\tan \gamma - 1)^2$$

Part E

$$\frac{\sin^2 x + 2 \sin x \cos x + \cos^2 x}{\sin x + \cos x} = \frac{(\sin x + \cos x)^2}{\sin x + \cos x} = \sin x + \cos x$$

Part F

$$\frac{\sin^2 x - 2 \sin x \cos x + \cos^2 x}{\sin x - \cos x} = \frac{(\sin x - \cos x)^2}{\sin x - \cos x} = \sin x - \cos x$$

Part G

Substitute $1 = \sin^2 \alpha + \cos^2 \alpha$:

$$\frac{\sin^2 \alpha + 2 \sin \alpha \cos \alpha + \cos^2 \alpha}{\sin \alpha + \cos \alpha} = \frac{(\sin \alpha + \cos \alpha)^2}{\sin \alpha + \cos \alpha} = \sin \alpha + \cos \alpha$$

Part H

Substitute $1 = \sin^2 \alpha + \cos^2 \alpha$:

$$\frac{\sin^2 \alpha - 2 \sin \alpha \cos \alpha + \cos^2 \alpha}{\sin \alpha - \cos \alpha} = \frac{(\sin \alpha - \cos \alpha)^2}{\sin \alpha - \cos \alpha} = \sin \alpha - \cos \alpha$$

Example 1.10

$$a^2 - b^2 = (a + b)(a - b)$$

Do as directed in the questions below. You might find the identities above useful.

Multiply and simplify:

- A. $(\sin \theta + \cos \theta)(\sin \theta - \cos \theta)$
- B. $(1 + \sin x)(1 - \sin x) + (1 + \cos x)(1 - \cos x)$

Factor and Simplify

- C. $\sin^4 \phi - \cos^4 \phi$
- D. $\sin^8 \phi - \cos^8 \phi$
- E. $\frac{\sin^2 x - \cos^2 x}{\sin x - \cos x}$
- F. $\frac{\sin^2 \phi - \cos^2 \phi}{\sin \phi + \cos \phi}$
- G. $\frac{\sin^4 \beta - \cos^4 \beta}{\sin \beta + \cos \beta}$

Part A

$$\begin{aligned} &= \sin^2 \theta - \cos^2 \theta \\ &= \sin^2 \theta + \underbrace{\cos^2 \theta - \cos^2 \theta}_{\text{Equal to Zero}} - \cos^2 \theta \\ &= 1 - 2 \cos^2 \theta \end{aligned}$$

Part B

Multiply:

$$1 - \sin^2 x + 1 - \cos^2 x = 2 - (\sin^2 x + \cos^2 x)$$

Substitute $\sin^2 x + \cos^2 x = 1$:

$$= 2 - 1 = 1$$

Part C

Factor using difference of squares:

$$(\sin^2 \phi + \cos^2 \phi)(\sin^2 \phi - \cos^2 \phi)$$

Substitute $\sin^2 \phi + \cos^2 \phi = 1$:

$$\sin^2 \phi - \cos^2 \phi$$

Factor again using difference of squares:

$$(\sin \phi + \cos \phi)(\sin \phi - \cos \phi)$$

Part D

$$(\sin^4 \phi + \cos^4 \phi)(\sin^4 \phi - \cos^4 \phi)$$

$$= (\sin^4 \phi + \cos^4 \phi)(\sin \phi + \cos \phi)(\sin \phi - \cos \phi)$$

Part E

$$\frac{\sin^2 x - \cos^2 x}{\sin x - \cos x} = \frac{(\sin x + \cos x)(\sin x - \cos x)}{\sin x - \cos x} = \sin x + \cos x$$

Part F

$$\frac{(\sin \phi + \cos \phi)(\sin \phi - \cos \phi)}{\sin \phi + \cos \phi} = \sin \phi - \cos \phi$$

Part G

$$\frac{(\sin^2 \beta - \cos^2 \beta)(\sin^2 \beta + \cos^2 \beta)}{\sin \beta + \cos \beta}$$

Substitute $1 = \sin^2 \beta + \cos^2 \beta$:

$$\frac{\sin^2 \beta - \cos^2 \beta}{\sin \beta + \cos \beta} = \frac{(\sin \beta + \cos \beta)(\sin \beta - \cos \beta)}{\sin \beta + \cos \beta} = \sin \beta - \cos \beta$$

Example 1.11

$$\begin{aligned} a^3 + b^3 &= (a + b)(a^2 - ab + b^2) \\ a^3 - b^3 &= (a - b)(a^2 + ab + b^2) \end{aligned}$$

Do as directed in the questions below. You might find the identities above useful.

Simplify:

- A. $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta}$
 B. $\frac{\sin^3 \gamma - \cos^3 \gamma}{1 - \sin^2 \gamma \cos^2 \gamma}$

Part A

Apply the formula for the sum of cubes:

$$\begin{aligned} &\frac{(\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)}{\sin \theta + \cos \theta} \\ &= \sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta \\ &1 - \sin \theta \cos \theta \end{aligned}$$

Part B

Apply the formula for the difference of cubes:

$$\begin{aligned} &\frac{(\sin \gamma - \cos \gamma)(\sin^2 \gamma + \sin \gamma \cos \gamma + \cos^2 \gamma)}{(1 - \sin \gamma \cos \gamma)(1 + \sin \gamma \cos \gamma)} \\ &= \frac{(\sin \gamma - \cos \gamma)(1 + \sin \gamma \cos \gamma)}{(1 - \sin \gamma \cos \gamma)(1 + \sin \gamma \cos \gamma)} \\ &= \frac{\sin \gamma - \cos \gamma}{1 - \sin \gamma \cos \gamma} \end{aligned}$$

Example 1.12

- A. $\sin^2 \theta + 5 \sin \theta + 6$
 B. $2 \cos^2 \alpha + \cos \alpha - 3$
 C.

Part A

Let $\sin \theta = a$

$$\begin{aligned} &a^2 + 5a + 6 \\ &(a + 2)(a + 3) \end{aligned}$$

Change back to the original variable:

$$(\sin \theta + 2)(\sin \theta + 3)$$

Part B

Use a change of variable. Let $\cos \alpha = a$:

$$2a^2 + a - 3$$

$$\text{Product} = -6 = (3)(-2), \text{Sum} = 1$$

$$\begin{aligned} &2a^2 + 3a - 2a - 3 \\ &a(2a + 3) - 1(2a + 3) \\ &(2a + 3)(a - 1) \end{aligned}$$

Change back to the original variable:

$$(2 \cos \alpha + 3)(\cos \alpha - 1)$$

C. Rationalization**1.13: Non-negative Square Roots**

We will make the assumption that all quantities inside a square root sign are positive. If we remove this assumption, we will have to redo all the questions that are given below.

Example 1.14

$$\sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}$$

Method I

$$= \frac{\sqrt{1 - \cos \alpha}}{\sqrt{1 + \cos \alpha}} \times \frac{\sqrt{1 + \cos \alpha}}{\sqrt{1 + \cos \alpha}} = \frac{\sqrt{1 - \cos^2 \alpha}}{1 + \cos \alpha} = \frac{\sqrt{\sin^2 \alpha}}{1 + \cos \alpha} = \frac{\sin \alpha}{1 + \cos \alpha}$$

Method II

$$= \frac{\sqrt{1 - \cos \alpha}}{\sqrt{1 + \cos \alpha}} \times \frac{\sqrt{1 - \cos \alpha}}{\sqrt{1 - \cos \alpha}} = \frac{1 - \cos \alpha}{\sqrt{1 - \cos^2 \alpha}} = \frac{1 - \cos \alpha}{\sqrt{\sin^2 \alpha}} = \frac{1 - \cos \alpha}{\sin \alpha}$$

Show that the two answers are the same.

$$\frac{\sin \alpha}{1 + \cos \alpha} \times \frac{1 - \cos \alpha}{1 - \cos \alpha} = \frac{(\sin \alpha)(1 - \cos \alpha)}{1 - \cos^2 \alpha} = \frac{(\sin \alpha)(1 - \cos \alpha)}{\sin^2 \alpha} = \frac{1 - \cos \alpha}{\sin \alpha}$$

Example 1.15

$$\sqrt{\sin^2 \alpha}$$

$$\sqrt{\sin^2 \alpha} = |\sin \alpha|$$

$$\begin{aligned} \sqrt{x^2} &= |x| \\ x > 0: \sqrt{x^2} &= x \\ x < 0: \sqrt{x^2} &= -x \end{aligned}$$

For Quadrant IV and I:

$$\cos x > 0: \sqrt{(1 + \sin x)(1 - \sin x)} = \sqrt{1 - \sin^2 x} = \sqrt{\cos^2 x} = \cos x$$

For Quadrant II and III:

$$\cos x < 0: \sqrt{(1 + \sin x)(1 - \sin x)} = \sqrt{1 - \sin^2 x} = \sqrt{\cos^2 x} = -\cos x$$

For all Quadrants:

$$\sqrt{(1 + \sin x)(1 - \sin x)} = \sqrt{1 - \sin^2 x} = \sqrt{\cos^2 x} = |\cos x|$$

Example 1.16

When $\alpha = 180^\circ$, determine the value of

$$\sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}$$

Substitute $\cos 180^\circ = -1$

$$\sqrt{\frac{1 - (-1)}{1 + (-1)}} = \sqrt{\frac{2}{0}} \Rightarrow \text{Not Defined}$$

D. Identities Derived from Pythagorean

1.17: Identities derived from the Pythagorean Identity

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Divide both sides of the Pythagorean Identity by $\cos^2 \theta$:

$$\sin^2 \theta + \cos^2 \theta = 1 \Rightarrow \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta} \Rightarrow \tan^2 \theta + 1 = \sec^2 \theta$$

Divide both sides of the Pythagorean Identity by $\sin^2 \theta$:

$$\sin^2 \theta + \cos^2 \theta = 1 \Rightarrow \frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta} \Rightarrow 1 + \cot^2 \theta = \csc^2 \theta$$

Example 1.18

$$(\sec x + \tan x)(\sec x - \tan x)$$

$$= \sec^2 x - \tan^2 x = \tan^2 x + 1 - \tan^2 x = 1$$

E. Proving Identities

There are three main ways you can work with and prove an identity

- Work with the LHS, and show that it is equal to the RHS
- Work with the RHS and show that it is equal to the LHS
- Work with both sides and show that they are equal

Example 1.19

Prove that:

$$\cos^2 x = \left(\frac{\csc x - 1}{\csc x} \right) \left(1 + \frac{1}{\csc x} \right)$$

Work with the RHS. Substitute $\csc x = \frac{1}{\sin x}$:

$$\begin{aligned} RHS &= \left(\frac{\frac{1}{\sin x} - 1}{\frac{1}{\sin x}} \right) (1 + \sin x) \\ &= \left(\frac{1 - \sin x}{\sin x} \right) \left(\frac{\sin x}{1} \right) (1 + \sin x) \\ &= 1 - \sin^2 x \\ &= \cos^2 x = LHS \end{aligned}$$

Example 1.20: Multiply by One

Show that:

$$\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta}$$

$$RHS = \frac{1}{\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}} = \frac{1}{\frac{1 - \sin \theta}{\cos \theta}}$$

Move the fraction in the denominator to the numerator by taking the reciprocal:

$$\frac{\cos \theta}{1 - \sin \theta}$$

We do not want a denominator, so multiply the numerator and the denominator by $1 + \sin \theta$:

$$\frac{\cos \theta (1 + \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)} = \frac{\cos \theta + \cos \theta \sin \theta}{\underbrace{1 - \sin^2 \theta}_{\because \cos^2 \theta = 1 - \sin^2 \theta}}$$

Substitute $\cos^2 \theta = 1 - \sin^2 \theta$

$$\frac{\cos \theta (1 + \sin \theta)}{\cos^2 \theta} = \frac{1 + \sin \theta}{\cos \theta} = \sec \theta + \tan \theta = LHS$$

Example 1.21

Consider the following set of steps used to show that $\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta}$

Start with the identity to be proved:

$$\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta}$$

Cross-multiply and expand:

$$\begin{aligned} (\sec \theta - \tan \theta)(\sec \theta + \tan \theta) &= 1 \\ \sec^2 \theta - \tan^2 \theta &= 1 \end{aligned}$$

Expand using the definition:

$$\frac{1}{\cos^2 \theta} - \frac{\sin^2 \theta}{\cos^2 \theta} = 1$$

Multiply both sides by $\cos^2 \theta$:

$$1 - \sin^2 \theta = \cos^2 \theta$$

Re-arrange:

$$\cos^2 \theta + \sin^2 \theta = 1$$

The above is a known identity.

Hence, the proof is complete.

Are the steps correct? If the steps are not correct, what can you do to fix it.

Each of the steps used above is reversible, and hence, the identity is correct.

Example 1.22

$$\text{Show that } \csc a - \cot a = \frac{1}{\csc a + \cot a}$$

$$LHS = \frac{1 - \cos a}{\sin a} = \frac{1}{\frac{\sin a (1 + \cos a)}{(1 - \cos a)(1 + \cos a)}} = \frac{1}{\frac{\sin a + \sin a \cos a}{\sin^2 a}} = \frac{1}{\frac{1}{\sin a} + \frac{\cos a}{\sin a}} = \frac{1}{\csc a + \cot a}$$

Example 1.23: Nested Fractions

$$\text{Show that } \frac{\sec x + \csc x}{\tan x + \cot x} = \sin x + \cos x$$

We start with the left-hand side since it is more complicated:

$$LHS = \frac{\sec x + \csc x}{\tan x + \cot x} = \frac{\frac{1}{\cos x} + \frac{1}{\sin x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}} = \frac{\frac{\sin x + \cos x}{\sin x \cos x}}{\frac{\sin^2 x + \cos^2 x}{\sin x \cos x}} = \sin x + \cos x = RHS$$

Example 1.24

$$(\sin x \tan x)^2 = (\tan x + \sin x)(\tan x - \sin x)$$

$$RHS = \tan^2 x - \sin^2 x = \frac{\sin^2 x}{\cos^2 x} - \sin^2 x = \sin^2 x \left(\frac{1}{\cos^2 x} - 1 \right) = \sin^2 x (\sec^2 x - 1) = (\sin x \tan x)^2$$

Example 1.25

$$\frac{\sin^2 x - \cos^2 x}{1 + 2 \sin x \cos x} = \frac{\tan x - 1}{\tan x + 1}$$

$$RHS = \frac{\tan x - 1}{\tan x + 1}$$

Substitute $\tan x = \frac{\sin x}{\cos x}$ and simplify:

$$= \frac{\frac{\sin x}{\cos x} - 1}{\frac{\sin x}{\cos x} + 1} = \frac{\frac{\sin x - \cos x}{\cos x}}{\frac{\sin x + \cos x}{\cos x}} = \frac{\sin x - \cos x}{\sin x + \cos x}$$

Note that on the LHS we have $\sin^2 x - \cos^2 x$. Hence, multiply by $\frac{\sin x + \cos x}{\sin x + \cos x}$:

$$= \frac{\sin x - \cos x}{\sin x + \cos x} \cdot \frac{\sin x + \cos x}{\sin x + \cos x} = \frac{\sin^2 x - \cos^2 x}{(\sin x + \cos x)^2}$$

Expand:

$$= \frac{\sin^2 x - \cos^2 x}{\sin^2 x + 2 \sin x \cos x + \cos^2 x}$$

Substitute $\sin^2 x + \cos^2 x = 1$:

$$= \frac{\sin^2 x - \cos^2 x}{1 + 2 \sin x \cos x} = LHS$$

F. Higher Powers

Example 1.26

Let $f_n(x) = \sin^n x + \cos^n x$. For how many x in $[0, \pi]$ is it true that $6f_4(x) - 4f_6(x) = 2f_2(x)$. (AMC 12P 2002/10)

Divide both sides of the given equation by 2:

$$3f_4(x) - 2f_6(x) = f_2(x)$$

Consider the RHS:

$$f_2(x) = \sin^2 x + \cos^2 x = 1$$

Consider the first term on the LHS:

$$3f_4(x) = 3(\sin^4 x + \cos^4 x)$$

Use a change of variable. Substitute $p = \sin x, q = \cos x$

$$\begin{aligned} & 3(p^4 + q^4) \\ &= 3(p^4 + 2p^2q^2 + q^4 - 2p^2q^2) \\ &= 3[(p^2 + q^2)^2 - 2p^2q^2] \end{aligned}$$

Substitute $p^2 + q^2 = \sin^2 x + \cos^2 x = 1$

$$\begin{aligned} &= 3[1 - 2p^2q^2] \\ &= 3 - 6p^2q^2 \end{aligned}$$

Consider the second term on the LHS:

$$2f_6(x) = 2(\sin^6 x + \cos^6 x)$$

Use a change of variable. Substitute $p = \sin x, q = \cos x$

$$\begin{aligned} & 2(p^6 + q^6) \\ &= 2(p^2 + q^2)(p^4 - p^2q^2 + q^4) \\ &= 2(p^4 - p^2q^2 + q^4) \\ &= 2(p^4 + 2p^2q^2 + q^4 - 3p^2q^2) \\ &= 2[(p^2 + q^2)^2 - 3p^2q^2] \\ &= 2[1 - 3p^2q^2] \\ &= 2 - 6p^2q^2 \end{aligned}$$

Therefore,

$$3f_4(x) - 2f_6(x) = 3 - 6p^2q^2 - (2 - 6p^2q^2) = 1$$

Since

$$\begin{aligned} LHS &= RHS \Rightarrow x \in \mathbb{R} \\ \text{Infinite Solutions} \end{aligned}$$

Example 1.27

Let $f(x) = \sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x}$. An equivalent form of $f(x)$ is: (AMC 12P 2002/17)

- A. $1 - \sqrt{2} \sin x$
- B. $-1 + \sqrt{2} \cos x$
- C. $\cos \frac{x}{2} - \sin \frac{x}{2}$

- D. $\cos x - \sin x$
E. $\cos 2x$

Substitute $\cos^2 x = 1 - \sin^2 x$, $\sin^2 x = 1 - \cos^2 x$ to convert each radical into an expression of only $\sin$ or only $\cos$:

$$\begin{aligned} & \sqrt{\sin^4 x + 4(1 - \sin^2 x)} - \sqrt{\cos^4 x + 4(1 - \cos^2 x)} \\ &= \sqrt{\sin^4 x - 4\sin^2 x + 4} - \sqrt{\cos^4 x - 4\cos^2 x + 4} \\ &= \sqrt{(\sin^2 x - 2)^2} - \sqrt{(\cos^2 x - 2)^2} \end{aligned}$$

$$\begin{aligned} \sin^2 x &< 1 \\ \sin^2 x - 2 &< -1 \end{aligned}$$

Hence, we must take the square roots as:

$$\begin{aligned} & (2 - \sin^2 x) - (2 - \cos^2 x) \\ &= \cos^2 x - \sin^2 x = \cos 2x \\ & \text{Option E} \end{aligned}$$

G. Application: Trig Substitutions

The Pythagorean identities are useful in eliminating square roots from algebraic expressions using a change of variable. This technique is exploited in Calculus to replace an expression that has a square root with one that doesn't.

1.28: Substitutions

Show that the following substitutions work:

- A. $x = a \sin \theta \Rightarrow \sqrt{a^2 - x^2} = a \cos \theta$
B. $x = a \tan \theta \Rightarrow \sqrt{a^2 + x^2} = a \sec \theta$
C. $x = a \sec \theta \Rightarrow \sqrt{x^2 - a^2} = a \tan \theta$

Part A

$$\sqrt{a^2 - x^2}$$

Substitute $x = a \sin \theta$:

$$\sqrt{a^2 - a^2 \sin^2 \theta} = \sqrt{a^2(1 - \sin^2 \theta)} = a\sqrt{1 - \sin^2 \theta}$$

Substitute $1 - \sin^2 \theta = \cos^2 \theta$:

$$a\sqrt{\cos^2 \theta} = a \cos \theta^1$$

Part B

$$\sqrt{a^2 + x^2}$$

Substitute $x = a \tan \theta$:

$$\sqrt{a^2 + a^2 \tan^2 \theta} = \sqrt{a^2(1 + \tan^2 \theta)} = a\sqrt{1 + \tan^2 \theta}$$

Here we make use of $\tan^2 \theta + 1 = \sec^2 \theta$:

$$= a\sqrt{\sec^2 \theta} = a \sec \theta$$

Part C

$$\sqrt{x^2 - a^2}$$

Substitute $x = a \sec \theta$:

$$\sqrt{a^2 \sec^2 \theta - a^2} = \sqrt{a^2(\sec^2 \theta - 1)} = a\sqrt{\sec^2 \theta - 1}$$

¹ Specifically $\sqrt{x^2} = |x|$. For the time being, let us assume that $\cos \theta$ is positive.

Here we make use of $\tan^2 \theta + 1 = \sec^2 \theta$:

$$= a\sqrt{\tan^2 \theta} = a \tan \theta$$

Example 1.29

Use a trigonometric substitution in terms of θ to remove the square root, and find the values of $\sin \theta$, $\cos \theta$, and $\tan \theta$ in terms of x .

- A. $\sqrt{1-x^2}$
- B. $\frac{x^2}{\sqrt{25-x^2}}$
- C. $\sqrt{4+x^2}$
- D. $\frac{1}{x^2\sqrt{16+x^2}}$
- E. $\sqrt{x^2-100}$
- F. $\sqrt{3-x^2}$
- G. $\sqrt{16-9x^2}$

Part A

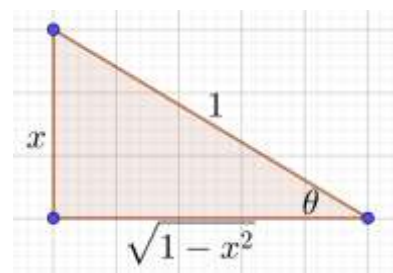
Substitute $x = \sin \theta$:

$$\sqrt{1-x^2} = \sqrt{1-\sin^2 \theta} = \sqrt{\cos^2 \theta} = \cos \theta$$

$$\sin \theta = \frac{x}{1}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \sqrt{1-x^2}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{x}{\sqrt{1-x^2}}$$



Part B

$$\frac{x^2}{\sqrt{25-x^2}}$$

Substitute $x = 5 \sin \theta$:

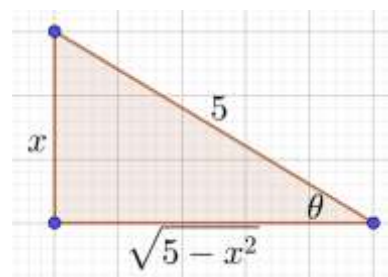
$$\frac{(5 \sin \theta)^2}{\sqrt{25-(5 \sin \theta)^2}} = \frac{25 \sin^2 \theta}{\sqrt{25-25 \sin^2 \theta}} = \frac{25 \sin^2 \theta}{5\sqrt{\cos^2 \theta}} = \frac{5 \sin^2 \theta}{\cos \theta}$$

$$x = 5 \sin \theta \Rightarrow \sin \theta = \frac{x}{5}$$

$$x^2 + \text{adj}^2 = 25 \Rightarrow \text{adj} = \sqrt{25-x^2}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{\sqrt{25-x^2}}{5}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{x}{\sqrt{25-x^2}}$$



Part C

Use a change of variable. Let $x = 2 \tan \theta$:

$$\sqrt{(4+4 \tan^2 \theta)} = \sqrt{(4+4 \tan^2 \theta)} = \sqrt{4(1+\tan^2 \theta)} = 2\sqrt{\sec^2 \theta} = 2 \sec \theta$$

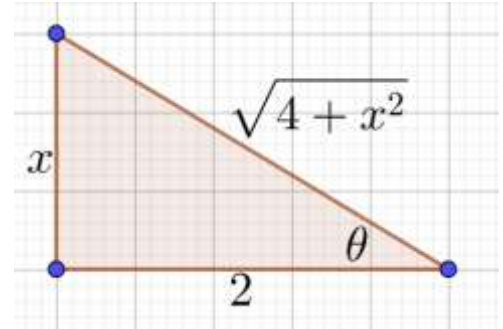
$$x = 2 \tan \theta \Rightarrow \tan \theta = \frac{x}{2}$$

Draw a reference triangle

$$\text{Hypotenuse} = \sqrt{4 + x^2}$$

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{x}{\sqrt{4 + x^2}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{2}{\sqrt{4 + x^2}}$$



Part D

Part E

Part F

Part G

$$\sqrt{16 - 9x^2} = \sqrt{9\left(\frac{16}{9} - x^2\right)}$$

Substitute $x = \frac{3}{2} \sin \theta$:

Example 1.30: Completing the square

- Complete the square below.
- Use a change of variable to bring it in line with the formula above.
- Use a trigonometric substitution in terms of θ to remove the square root, and find the values of $\sin \theta$, $\cos \theta$, and

$\tan \theta$.

$$\sqrt{-x^2 + 2x + 5}$$

Part A

$$\sqrt{-x^2 + 2x + 5} = \sqrt{4 - (x - 1)^2}$$

Part B

Let $u = x - 1$:

$$\sqrt{4 - u^2}$$

Part C

Let $u = \sin \theta$

Example 1.31

$$\sqrt{4x^2 + 4x - 48}$$

- Complete the square.

- B. Use a change of variable to bring it in line with the formula above.
- C. Factor the expression inside the square root to make the coefficient of x as 1.
- D. Use a trigonometric substitution in terms of θ to remove the square root, and find the values of $\sin \theta$, $\cos \theta$, and $\tan \theta$.

H. Expressions of power $\frac{n}{2}$

Example 1.32

I. Trigonometric Expressions

If the expression inside the square root is itself trigonometric, you can still use a change of variable to eliminate the square root.

Example 1.33

Use a trigonometric substitution in terms of θ to remove the square root, and find the values of $\sin \theta$, $\cos \theta$, and $\tan \theta$.

$$\sqrt{1 + \sin^2 x}$$

1.3 Double Angle and Half Angle Identities

A. Double Angle Identities for $\sin$

1.34: Double Angle Identity

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

Use $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$:

$$\sin(2\theta) = \sin(\theta + \theta) = \sin \theta \cos \theta + \cos \theta \sin \theta = 2 \sin \theta \cos \theta$$

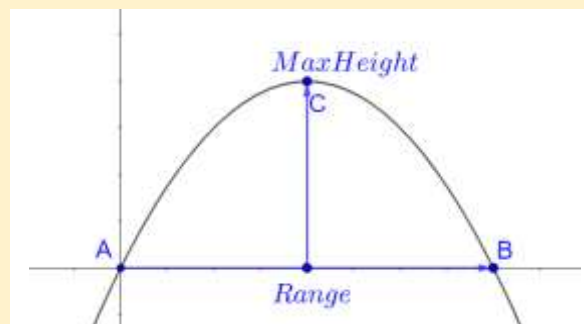
Example 1.35: Physics

The range of a projectile is given by $R = u_x t + \frac{1}{2} a_x t^2$, where the displacement $= u_x = u \cos \theta$, the flight is given by $t = \frac{2u \sin \theta}{g}$, the acceleration

$$\begin{aligned} \text{Acceleration} &= a_x = 0 \\ \theta &= \text{Launch Angle} \end{aligned}$$

Show that

$$\frac{u^2 \sin 2\theta}{g}$$



Substitute the given values in the given equation:

$$R = u_x t + \frac{1}{2} a_x t^2 = (u \cos \theta) \left(\frac{2u \sin \theta}{g} \right) + \frac{1}{2} (0) t^2$$

The second term will become zero, and carry out the multiplication in the first term:

$$= \frac{u^2(2 \sin \theta \cos \theta)}{g}$$

Substitute $2 \sin \theta \cos \theta = \sin 2\theta$ using the double angle formula:

$$= \frac{u^2 \sin 2\theta}{g}$$

Example 1.36: Geometry

In a right triangle the square of the hypotenuse is equal to twice the product of the legs. One of the acute angles of the triangle is: (AHSME 1959/15)

Method I: Trigonometry

$$h^2 = 2l_1l_2$$

Substitute $\sin \theta = \frac{l_1}{h} \Rightarrow l_1 = h \sin \theta$, $\cos \theta = \frac{l_2}{h} \Rightarrow l_2 = h \cos \theta$:

$$h^2 = 2(h \sin \theta)(h \cos \theta)$$

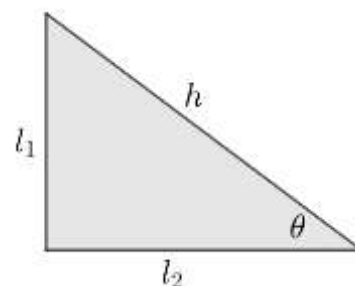
$$1 = 2 \sin \theta \cos \theta$$

Using the double angle identity:

$$\sin 2\theta = 1$$

Solving for θ

$$2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$



Method II: Pythagoras Theorem

Let the legs be x and y . Then:

$$x^2 + y^2 = 2xy \Rightarrow (x - y)^2 = 0 \Rightarrow x - y = 0 \Rightarrow x = y$$

Hence, the triangle is an isosceles triangle. And we know that it is right-angled. Combine the two to get:

$$\text{Right-angled isosceles triangle} \Rightarrow 45 - 45 - 90$$

Example 1.37

Prove that

$$\sin 2\theta = (\sin \theta + \cos \theta)^2 + \tan^2 \theta - \sec^2 \theta$$

Start with the RHS. Substitute $\tan^2 \theta - \sec^2 \theta = -1$:

$$(\sin \theta + \cos \theta)^2 - 1$$

Open the brackets:

$$\sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta - 1$$

Substitute $\sin^2 \theta + \cos^2 \theta = 1$

$$2 \sin \theta \cos \theta + 1 - 1$$

$$= 2 \sin \theta \cos \theta$$

By the double angle identity for sine:

$$= \sin 2\theta = LHS$$

Example 1.38: Telescoping

The value of the expression below as a single real number: (JEE-M 2019)

$$\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \dots \cdot \cos \frac{\pi}{2^{10}} \cdot \sin \frac{\pi}{2^{10}}$$

$$\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \dots \cdot \cos \frac{\pi}{2^9} \cdot \cos \frac{\pi}{2^{10}} \cdot \sin \frac{\pi}{2^{10}}$$

Use the double angle identity $\frac{1}{2} \cdot \sin(2\theta) = \sin \theta \cos \theta$ on the last two terms:

$$\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \dots \cdot \cos \frac{\pi}{2^9} \cdot \left(\frac{1}{2} \cdot \sin \frac{\pi}{2^9} \right)$$

Use the double angle identity $\frac{\sin(2\theta)}{2} = \sin \theta \cos \theta$ again on the last two terms:

$$\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \dots \cdot \cos \frac{\pi}{2^8} \cdot \left(\frac{1}{2^2} \cdot \sin \frac{\pi}{2^8} \right)$$

Keep repeating. There are a total of ten terms, which combine nine to get:

$$\frac{1}{2^9} \sin \frac{\pi}{2} = \frac{1}{2^9} \cdot 1 = \frac{1}{512}$$

1.39: Telescoping Identity

$$\cos A \cdot \cos 2A \cdot \cos 2^2 A \cdot \dots \cdot \cos 2^{n-1} A = \frac{\sin 2^n A}{2^n \sin A}$$

$$LHS = \frac{2 \sin A}{2 \sin A} \times \cos A \cdot \cos 2A \cdot \cos 2^2 A \cdot \dots \cdot \cos 2^{n-1} A$$

$$= \frac{2 \sin 2A}{2^2 \sin A} \cdot \cos 2A \cdot \cos 2^2 A \cdot \dots \cdot \cos 2^{n-1} A$$

$$= \frac{2 \sin 2^2 A}{2^3 \sin A} \cdot \cos 2^2 A \cdot \dots \cdot \cos 2^{n-1} A$$

Keep repeating the process to telescope:

$$= \frac{\sin 2^n A}{2^n \sin A} = RHS$$

B. Double Angle Identities for $\cos$

1.40: Double Angle Identity

The double angle identity for $\cos 2\theta$ has three versions, all of which are important.

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$$

Use $\cos(a + b) = \cos a \cos b - \sin a \sin b$:

$$\cos 2\theta = \cos(\theta + \theta) = \cos \theta \cos \theta - \sin \theta \sin \theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = (1 - \sin^2 \theta) - \sin^2 \theta = \cos^2 \theta - (1 - \cos^2 \theta) = 2 \cos^2 \theta - 1$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$2 \sin^2 \theta = 1 - \cos 2\theta$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\begin{aligned}\cos 2\theta &= 2 \cos^2 \theta - 1 \\ 2 \cos^2 \theta &= 1 + \cos 2\theta \\ \cos^2 \theta &= \frac{1 + \cos 2\theta}{2}\end{aligned}$$

Example 1.41

$$\frac{\cos 2x}{\cos x} - \frac{\sin 2x}{\sin x} = -\sec x$$

Use the double angle identities in the LHS:

$$\begin{aligned}&= \frac{\cos^2 x - \sin^2 x}{\cos x} - \frac{2 \sin x \cos x}{\sin x} \\ &= \frac{\cos^2 x - \sin^2 x}{\cos x} - 2 \cos x\end{aligned}$$

Add the fractions:

$$= \frac{\cos^2 x - \sin^2 x - 2 \cos^2 x}{\cos x}$$

Simplify:

$$= \frac{-(\sin^2 x + \cos^2 x)}{\cos x}$$

Substitute $\sin^2 x + \cos^2 x = 1$

$$= -\sec x = RHS$$

1.42: Reduction of Higher Powers

The double angle identities can be used to “reduce” the powers of trigonometric functions by writing them in lower powers.

Challenge 1.43

Reduce the LHS to show that it is equal to the RHS:

$$\sin^4 x = \frac{3 - 4 \cos 2x + \cos 4x}{8}$$

$$LHS = \sin^4 x$$

Substitute $\sin^4 x = (\sin^2 x)^2$:

$$= (\sin^2 x)^2$$

$$\begin{aligned}\text{Substitute } \cos 2\theta &= 1 - 2 \sin^2 \theta \Rightarrow 2 \sin^2 \theta = 1 - \cos 2\theta \Rightarrow \sin^2 \theta = \frac{1 - \cos 2\theta}{2} \\ &\left(\frac{1 - \cos 2x}{2} \right)^2\end{aligned}$$

Expand using $(a - b)^2 = a^2 - 2ab + b^2$

$$= \frac{1 - 2 \cos 2x + \cos^2 2x}{4}$$

Substitute $\cos 2x = 2 \cos^2 x - 1 \Rightarrow \cos^2 x = \frac{1 + \cos 2x}{2}$:

$$\frac{1 - 2 \cos 2x + \left(\frac{1 + \cos 4x}{2} \right)}{4}$$

Simplify:

$$= \frac{3 - 4 \cos 2x + \cos 4x}{8}$$

1.44: Finding values of trigonometric functions

The double angle identities can be used to find the values of certain important trigonometric functions.

Example 1.45

Find $\cos 15^\circ$

$$\cos x = \sqrt{\frac{1 + \cos 2x}{2}}$$

Substitute $x = 15^\circ$

$$\cos 15^\circ = \sqrt{\frac{1}{2}(\cos 30 + 1)} = \sqrt{\frac{1}{2}\left(\frac{\sqrt{3}}{2} + 1\right)} = \sqrt{\left(\frac{\sqrt{3} + 2}{4}\right)} = \frac{1}{2}\sqrt{\sqrt{3} + 2}$$

1.46: Identity

$$\frac{1 + \cos \theta}{2} = \cos^2 \frac{\theta}{2}$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$1 + \cos 2\alpha = 2 \cos^2 \alpha$$

$$1 + \cos 2\alpha = 2 \cos^2 \alpha$$

$$\text{Let } 2\alpha = \theta \Rightarrow \alpha = \frac{\theta}{2}$$

$$1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$$

Example 1.47

Simplify the expression below and write in the form $a \cos \alpha$:

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos \theta}}}$$

Factor 2 from the innermost root:

$$\sqrt{2 + \sqrt{2 + \sqrt{2(1 + \cos \theta)}}}$$

Substitute $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$:

$$= \sqrt{2 + \sqrt{2 + \sqrt{2\left(2 \cos^2 \frac{\theta}{2}\right)}}} = \sqrt{2 + \sqrt{2 + 2 \cos \frac{\theta}{2}}}$$

Repeat the process:

$$= \sqrt{2 + \sqrt{2 \left(1 + 1 \cos \frac{\theta}{2}\right)}} = \sqrt{2 + \sqrt{2 \left(2 \cos^2 \frac{\theta}{4}\right)}} = \sqrt{2 + 2 \cos \frac{\theta}{4}}$$

Repeat the process one more time:

$$= \sqrt{2 \left(1 + \cos \frac{\theta}{4}\right)} = \sqrt{2 \left(2 \cos^2 \frac{\theta}{8}\right)} = 2 \cos \frac{\theta}{8}$$

1.48: Double Angle Identity for *tan*

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Example 1.49

$$\begin{aligned} \cos(2x + 2x) &= \cos^2 2x - \sin^2 2x = (\cos^2 x - \sin^2 x)^2 - (2 \sin x \cos x)^2 \\ &= \cos^4 x - 2 \cos^2 x \sin^2 x + \sin^4 x - 4 \sin^2 x \cos^2 x \\ &= \cos^4 x - 6 \cos^2 x \sin^2 x + \sin^4 x \\ &\quad \text{Version 1} \\ &= \cos^4 x + 2 \cos^2 x \sin^2 x + \sin^4 x - 8 \cos^2 x \sin^2 x \\ &= \cos^4 x + 2 \cos^2 x \sin^2 x + \sin^4 x - 8 \cos^2 x \sin^2 x \\ &= (\sin^2 x + \cos^2 x)^2 - 8 \cos^2 x \sin^2 x \\ &= 1 - 8 \cos^2 x \sin^2 x \\ &\quad \text{Version 2} \\ \cos 4x &= \cos(2x + 2x) = 2 \cos^2 2x - 1 = 2(2 \cos^2 x - 1)^2 - 1 \\ &= 2(4 \cos^4 x - 4 \cos^2 x + 1) - 1 = 8 \cos^4 x - 8 \cos^2 x + 1 \\ &\quad \text{Version 3} \end{aligned}$$

C. Half Angle Identities

The half angle identities are used to find calculate trigonometric functions for “half” of an angle for which you know the value.

1.50: $\sin \frac{\theta}{2}$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\begin{aligned} \cos 2\theta &= 1 - 2 \sin^2 \theta \\ 2 \sin^2 \theta &= 1 - \cos 2\theta \\ \sin^2 \theta &= \frac{1 - \cos 2\theta}{2} \\ \sin^2 \frac{\theta}{2} &= \frac{1 - \cos \theta}{2} \\ \sin \frac{\theta}{2} &= \pm \sqrt{\frac{1 - \cos \theta}{2}} \end{aligned}$$

1.51: $\cos \frac{\theta}{2}$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

Notes:

- When making use of this identity, the sign is determined by the quadrant in which the angle lies.
- $\cos \theta > 0 \Rightarrow \theta$ in *QI* or *QIV*

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$2 \cos^2 \theta = 1 + \cos 2\theta$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

Example 1.52Find the value of $\cos \frac{\pi}{8} = \cos 22.5^\circ$.

Note that

- $22.5 = \frac{45}{2}$, and hence we are in a position to make use of the half angle identity for $\cos$.
- $\cos \frac{\pi}{8}$ is in the first quadrant, and hence positive.

$$\cos \frac{\pi}{8} = \cos \left(\frac{\frac{\pi}{4}}{2} \right) = \sqrt{\frac{1 + \cos \frac{\pi}{4}}{2}} = \sqrt{\frac{1 + \frac{\sqrt{2}}{2}}{2}} = \sqrt{\left(\frac{2 + \sqrt{2}}{2} \right) \times \frac{1}{2}} = \sqrt{\frac{2 + \sqrt{2}}{4}} = \frac{1}{2} \sqrt{2 + \sqrt{2}}$$

1.53: $\tan \frac{\theta}{2}$

$$\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\tan \frac{\theta}{2} = \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} = \frac{\pm \sqrt{\frac{1 - \cos \theta}{2}}}{\pm \sqrt{\frac{1 + \cos \theta}{2}}} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$= \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta}} = \pm \sqrt{\frac{1 - \cos^2 \theta}{(1 + \cos \theta)^2}} = \pm \sqrt{\frac{\sin^2 \theta}{(1 + \cos \theta)^2}} = \pm \left| \frac{\sin \theta}{1 + \cos \theta} \right|$$

$$\sqrt{x^2} = |x|$$

$$\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$$

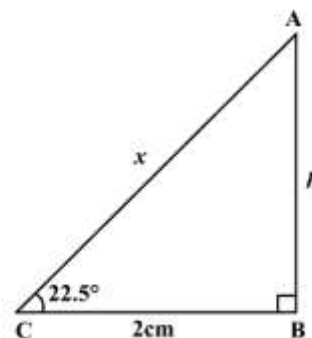
Example 1.54

A ladder kept in support of a wall makes an angle of 22.5° with the ground. The distance between the bottom of the wall and the foot of the ladder is 2. What is the length of the ladder? (JMET 2011/85)

Write your answer in the form $\sqrt{a + b\sqrt{c}}$, where $a, b, c \in \mathbb{N}$ and c has no perfect square factor.

$$\cos 22.5 = \frac{2}{x} \Rightarrow x = \frac{2}{\cos 22.5}$$

$$x = \frac{2}{\frac{1}{2}\sqrt{2+\sqrt{2}}} = \frac{4}{\sqrt{2+\sqrt{2}}} = \sqrt{\frac{16}{2+\sqrt{2}}} = \sqrt{8(2-\sqrt{2})} = \sqrt{16-8\sqrt{2}}$$



1.55: $\tan$

$$\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

1.56: Connecting cosines with Circumradius

If A, B and C are three angles in a triangle and $R = \text{circumradius}$ then:

$$\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{s}{4R}$$

Finding an expression for each term

Use the half-angle identity:

$$\cos \frac{A}{2} = \sqrt{\frac{\cos A + 1}{2}}$$

Substitute $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ and then simplify:

$$\sqrt{\frac{\frac{b^2 + c^2 - a^2}{2bc} + 1}{2}} = \sqrt{\frac{b^2 + c^2 + 2bc - a^2}{4bc}}$$

Factor the numerator, and then use difference of squares:

$$\sqrt{\frac{(b+c)^2 - a^2}{4bc}} = \sqrt{\frac{(a+b+c)(a+b+c-2a)}{4bc}}$$

Substitute $s = \frac{a+b+c}{2}$, and then simplify:

$$= \sqrt{\frac{(2s)(2s-2a)}{4bc}} = \sqrt{\frac{s(s-a)}{bc}}$$

Similarly, we can show that

$$\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}, \quad \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$$

Finding the product

$$\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \sqrt{\frac{s^3(s-a)(s-b)(s-c)}{a^2b^2c^2}} = \frac{s}{abc} \sqrt{s(s-a)(s-b)(s-c)}$$

Substitute $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ to get:

$$\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{s\Delta}{abc} = s \times \frac{\Delta}{abc}$$

And we know that $R = \frac{abc}{4\Delta} \Rightarrow 4R = \frac{abc}{\Delta} \Rightarrow \frac{\Delta}{abc} = \frac{1}{4R}$:

$$\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{s\Delta}{abc} = s \times \frac{1}{4R} = \frac{s}{4R}$$

1.57: Ratio of inradius and circumradius in terms of sines

$$\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{r}{4R}$$

In $\triangle IBD$:

$$\begin{aligned} \tan \frac{B}{2} &= \frac{ID}{BD} = \frac{r}{BD} \Rightarrow BD = r \cot \frac{B}{2} \\ \tan \frac{C}{2} &= \frac{ID}{BD} = \frac{r}{BD} \Rightarrow BD = r \cot \frac{C}{2} \end{aligned}$$

In $\triangle ABC$:

$$a = BC = BD + DC = r \cot \frac{B}{2} + r \cot \frac{C}{2} = r \left(\frac{\cos \frac{B}{2}}{\sin \frac{B}{2}} + \frac{\cos \frac{C}{2}}{\sin \frac{C}{2}} \right)$$

Add the fractions, and then use the double angle identity:

$$= r \left(\frac{\sin \frac{C}{2} \cos \frac{B}{2} + \sin \frac{B}{2} \cos \frac{C}{2}}{\sin \frac{B}{2} \sin \frac{C}{2}} \right) = r \frac{\sin \left(\frac{B}{2} + \frac{C}{2} \right)}{\sin \frac{B}{2} \sin \frac{C}{2}}$$

Since A, B and C are the angles of a triangle, we must have:

$$A + B + C = \pi \Rightarrow \frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2} \Rightarrow \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2} - \frac{A}{2}$$

Hence, we can substitute $\sin \left(\frac{B}{2} + \frac{C}{2} \right) = \sin \left(\frac{\pi}{2} - \frac{A}{2} \right) = \cos \frac{A}{2}$ to get:

$$a = r \frac{\cos \frac{A}{2}}{\sin \frac{B}{2} \sin \frac{C}{2}}$$

We can also substitute for a in terms of half-angles since $\frac{a}{\sin A} = 2R \Rightarrow a = 2R \sin A = 4R \sin \frac{A}{2} \cos \frac{A}{2}$:

$$4R \sin \frac{A}{2} \cos \frac{A}{2} = r \frac{\cos \frac{A}{2}}{\sin \frac{B}{2} \sin \frac{C}{2}} \Rightarrow \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{r}{4R}$$

1.58: Relation between inradius and circumradius

$$R \geq 2r$$

Using Jensen's Inequality, it can be shown that:

$$\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$$

And substitute $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{r}{4R}$

$$\frac{r}{4R} \leq \frac{1}{8} \Rightarrow R \geq 2r$$

1.4 Sum and Difference Identities

A. Calculating Values

1.59: *sin* and *cos*

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

Note that:

- Sine of a sum, RHS has a plus sign.
- Sine of a difference, RHS has a negative sign
- Cosine of a sum, RHS has a plus sign.
- Cosine of a difference, RHS has a negative sign

Example 1.60

Find the following:

- A. $\sin 75^\circ$
- B. $\sin 15^\circ$
- C. $\cos 75^\circ$
- D. $\cos 15^\circ$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin 75^\circ = \sin(45 + 30) = \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\sin 45} \underbrace{\left(\frac{\sqrt{3}}{2}\right)}_{\cos 30} + \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\cos 45} \underbrace{\left(\frac{1}{2}\right)}_{\sin 30} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

$$\sin(45 - 30) = \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\sin 45} \underbrace{\left(\frac{\sqrt{3}}{2}\right)}_{\cos 30} - \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\cos 45} \underbrace{\left(\frac{1}{2}\right)}_{\sin 30} = \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\cos(45 + 30) = \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\cos 45} \underbrace{\left(\frac{\sqrt{3}}{2}\right)}_{\cos 30} - \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\sin 45} \underbrace{\left(\frac{1}{2}\right)}_{\sin 30} = \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\cos(45 - 30) = \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\cos 45} \underbrace{\left(\frac{\sqrt{3}}{2}\right)}_{\cos 30} + \underbrace{\left(\frac{\sqrt{2}}{2}\right)}_{\sin 45} \underbrace{\left(\frac{1}{2}\right)}_{\sin 30} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

1.61: *tan*

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

Use the definition and then the formula:

$$\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}$$

Divide numerator and denominator by $\cos \alpha \cos \beta$, and simplify:

$$\begin{aligned} & \frac{\frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} + \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta}}{\frac{\cos \alpha \cos \beta}{\cos \alpha \cos \beta} - \frac{\sin \alpha \sin \beta}{\cos \alpha \cos \beta}} = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \end{aligned}$$

Example 1.62

- A. $\tan 75^\circ$
- B. $\tan 15^\circ$

$$\tan(45 + 30) = \frac{1 + \frac{\sqrt{3}}{3}}{1 - (1)\left(\frac{\sqrt{3}}{3}\right)} = \frac{\frac{3 + \sqrt{3}}{3}}{\frac{3 - \sqrt{3}}{3}} = \frac{3 + \sqrt{3}}{3 - \sqrt{3}} = \frac{(3 + \sqrt{3})^2}{6} = \frac{9 + 6\sqrt{3} + 3}{6} = 2 + \sqrt{3}$$

$$\tan(45 + 30) = \frac{1 + \frac{1}{\sqrt{3}}}{1 - (1)\left(\frac{1}{\sqrt{3}}\right)} = \frac{\frac{\sqrt{3} + 1}{\sqrt{3}}}{\frac{\sqrt{3} - 1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$\tan(45 - 30) = \frac{1 - \frac{1}{\sqrt{3}}}{1 + (1)\left(\frac{1}{\sqrt{3}}\right)} = \frac{\frac{\sqrt{3} - 1}{\sqrt{3}}}{\frac{\sqrt{3} + 1}{\sqrt{3}}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$\tan(45 - 30) = \frac{1 - \frac{\sqrt{3}}{3}}{1 + (1)\left(\frac{\sqrt{3}}{3}\right)} = \frac{\frac{3 - \sqrt{3}}{3}}{\frac{3 + \sqrt{3}}{3}} = \frac{3 - \sqrt{3}}{3 + \sqrt{3}} = \frac{(3 - \sqrt{3})^2}{6} = \frac{9 - 6\sqrt{3} + 3}{6} = 2 - \sqrt{3}$$

Example 1.63: Review

In $\triangle ABC$ is right-angled with $\angle A = \alpha$, $\angle B = 90^\circ$, $\angle C = \beta$, $AB = 5$, $BC = 12$. Find:

- A. $\sin(\alpha + \beta)$
- B. $\sin(\alpha - \beta)$
- C. $\cos(\alpha + \beta)$
- D. $\cos(\alpha - \beta)$
- E. $\tan(\alpha + \beta)$
- F. $\tan(\alpha - \beta)$

Example 1.64: Simple Harmonic Oscillator

In physics, a simple harmonic oscillator is used to model alternating current, the movement of an ideal spring and waves. The equation of a simple harmonic oscillator:

$$x = a \cos\left(\frac{2\pi}{T}t + \phi\right)$$

Where

$T = \text{Time period of oscillation}$

$\phi = \text{phase shift}$

- A. Use a change of variable. Substitute *angular frequency* $= \omega = \frac{2\pi}{T}$ to simplify the equation.
 B. Show that $(a \cos \phi)(\cos \omega t) - (a \sin \phi)(\sin \omega t)$ is also the equation of a simple harmonic oscillator.

$$x = a \cos(\omega t + \phi)$$

$$a[(\cos \phi)(\cos \omega t) - (\sin \phi)(\sin \omega t)] = a \cos(\omega t + \phi)$$

B. Identities

Example 1.65

$$\sin(90 - \theta) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

Using $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$:

$$LHS = \sin(90 - \theta) = \sin 90^\circ \cos \theta - \cos 90^\circ \sin \theta = (1) \cos \theta - (0) \sin \theta = \cos \theta = RHS$$

Using $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$:

$$LHS = \cos\left(\frac{\pi}{2} - \theta\right) = \cos\left(\frac{\pi}{2}\right) \cos(\theta) + \sin\left(\frac{\pi}{2}\right) \sin \theta = 0 \times \cos(\theta) + 1 \times \sin \theta = \sin \theta$$

Example 1.66

Prove the following identities:

$$A: \frac{\sin(\alpha + \beta)}{\sin(\alpha - \beta)} = \frac{\tan \alpha + \tan \beta}{\tan \alpha - \tan \beta}$$

$$B: \frac{\cos(\alpha + \beta)}{\cos(\alpha - \beta)} = \frac{1 - \tan \alpha \tan \beta}{1 + \tan \alpha \tan \beta}$$

Part A

Use the formula to expand the LHS and then divide by $\cos \alpha \cos \beta$ in the numerator and denominator:

$$LHS = \frac{(\sin \alpha \cos \beta + \cos \alpha \sin \beta) \times \frac{1}{\cos \alpha \cos \beta}}{(\sin \alpha \cos \beta - \cos \alpha \sin \beta) \times \frac{1}{\cos \alpha \cos \beta}} = \frac{\tan \alpha + \tan \beta}{\tan \alpha - \tan \beta} = RHS$$

Part B

Use the formula to expand the LHS, and since we want RHS to begin with a 1, divide by $\cos \alpha \cos \beta$ in the numerator and denominator:

$$LHS = \frac{(\cos \alpha \cos \beta - \sin \alpha \sin \beta) \times \frac{1}{\cos \alpha \cos \beta}}{(\cos \alpha \cos \beta + \sin \alpha \sin \beta) \times \frac{1}{\cos \alpha \cos \beta}} = \frac{1 - \tan \alpha \tan \beta}{1 + \tan \alpha \tan \beta} = RHS$$

Example 1.67

Identities like the ones below are useful in Calculus². Prove them:

$$A: \frac{\sin(x + h) - \sin x}{h} = \sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right)$$

$$B: \frac{\cos(x + h) - \cos x}{h} = \cos x \left(\frac{\cos h - 1}{h} \right) - \sin x \left(\frac{\sin h}{h} \right)$$

² For calculating derivatives of trigonometric functions using the limit definition.

Use the formula to expand, rearrange and factor, and then split the fraction:

$$A: LHS = \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} = \sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) = RHS$$

$$B: LHS = \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \cos x \left(\frac{\cos h - 1}{h} \right) - \sin x \left(\frac{\sin h}{h} \right) = RHS$$

Example 1.68

Prove:

$$\sin(a + b) \sin(a - b) = \sin^2 a - \sin^2 b$$

Use the sum and difference formula:

$$LHS = [\sin a \cos b + \sin b \cos a][\sin a \cos b - \sin b \cos a]$$

Use $(x + y)(x - y) = x^2 - y^2$:

$$\begin{aligned} & \sin^2 a \cos^2 b - \sin^2 b \cos^2 a \\ &= \sin^2 a (1 - \sin^2 b) - \sin^2 b (1 - \sin^2 a) \\ &= \sin^2 a - \sin^2 a \sin^2 b - \sin^2 b + \sin^2 a \sin^2 b \\ &= \sin^2 a - \sin^2 b = RHS \end{aligned}$$

Example 1.69

If $\cos(a + b) = \cos a \cos b$ show that:

$$\sin^2(a + b) = (\sin a + \sin b)^2$$

Expand the LHS using the sum formula for $\sin$:

$$(\sin a \cos b + \sin b \cos a)^2$$

Square:

$$= \sin^2 a \cos^2 b + 2 \sin a \sin b \cos a \cos b + \sin^2 b \cos^2 a$$

Substitute $\cos a \cos b - \sin a \sin b = \cos a \cos b \Rightarrow \sin a \sin b = 0$:

$$= \sin^2 a \cos^2 b + \sin^2 b \cos^2 a$$

Substitute $\cos^2 b = 1 - \sin^2 b$, $\cos^2 a = 1 - \sin^2 a$:

$$= \sin^2 a (1 - \sin^2 b) + \sin^2 b (1 - \sin^2 a)$$

Expand and simplify:

$$\begin{aligned} &= \sin^2 a - \sin^2 a \sin^2 b + \sin^2 b - \sin^2 a \sin^2 b \\ &= \sin^2 a + \sin^2 b \end{aligned}$$

Substitute $\sin a \sin b = 0$:

$$\begin{aligned} &= \sin^2 a + 2 \sin a \sin b + \sin^2 b \\ &= (\sin a + \sin b)^2 = RHS \end{aligned}$$

Example 1.70

Prove:

$$\frac{\sin(a - b)}{\sin a \sin b} + \frac{\sin(b - c)}{\sin b \sin c} + \frac{\sin(c - a)}{\sin c \sin a} = 0$$

Rearrange:

$$\begin{aligned} \frac{\sin(a-b)}{\sin a \sin b} + \frac{\sin(b-c)}{\sin b \sin c} &= -\frac{\sin(c-a)}{\sin c \sin a} \\ LHS &= \frac{\sin(a-b) \sin c + \sin(b-c) \sin a}{\sin a \sin b \sin c} \\ &= \frac{[\sin a \cos b - \sin b \cos a] \sin c + [\sin b \cos c - \sin c \cos b] \sin a}{\sin a \sin b \sin c} \\ &= \frac{[\sin a \cos b \sin c - \sin b \sin c \cos a] + [\sin a \sin b \cos c - \sin a \cos b \sin c]}{\sin a \sin b \sin c} \\ &= \frac{\sin a \sin b \sin c}{\sin a \sin b \sin c} = \frac{\sin a \sin b \cos c - \sin b \sin c \cos a}{\sin a \sin b \sin c} \\ &= \frac{\sin(a-c)}{\sin a \sin c} = \frac{-\sin(c-a)}{\sin a \sin c} = RHS \end{aligned}$$

C. Expressions

Example 1.71

Determine the value of $\cos(A+B)$ if

$$\sin\left(\frac{\pi}{2} - A\right) \sin\left(\frac{\pi}{2} - B\right) = \cos\left(\frac{\pi}{2} - A\right) \cos\left(\frac{\pi}{2} - B\right)$$

Using the cofunction identities $\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$, $\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$:

$$\begin{aligned} \cos A \cos B &= \sin A \sin B \\ \cos A \cos B - \sin A \sin B &= 0 \end{aligned}$$

Using the sum formula for $\cos$:

$$\cos(A+B) = 0$$

Example 1.72

Simplify

- A. $\cos(A+B) \cos B + \sin(A+B) \sin B$
- B. $\sin(A-B) \cos B + \cos(A-B) \sin B$

Part A

$$\begin{aligned} &= [\cos A \cos B - \sin A \sin B] \cos B + [\sin A \cos B + \sin B \cos A] \sin B \\ &= [\cos A \cos^2 B - \sin A \sin B \cos B] + [\sin A \sin B \cos B + \sin^2 B \cos A] \end{aligned}$$

Simplify, factor, and use the Pythagorean Identity:

$$\begin{aligned} &= \cos A \cos^2 B + \sin^2 B \cos A \\ &= \cos A (\cos^2 B + \sin^2 B) = \cos A \end{aligned}$$

Part B

$$\begin{aligned} &= [\sin A \cos B - \sin B \cos A] \cos B + [\cos A \cos B + \sin A \sin B] \sin B \\ &= [\sin A \cos^2 B - \sin B \cos A \cos B] + [\sin B \cos A \cos B + \sin A \sin^2 B] \\ &= \sin A \cos^2 B + \sin A \sin^2 B \\ &= \sin A \end{aligned}$$

Example 1.73

Simplify

$$\frac{\sin(A+B) - \sin(A-B)}{\cos(A+B) + \cos(A-B)}$$

Expand using the sum and difference formula:

$$= \frac{(\sin A \cos B + \sin B \cos A) - (\sin A \cos B - \sin B \cos A)}{(\cos A \cos B - \sin A \sin B) + (\cos A \cos B + \sin A \sin B)}$$

Simplify:

$$= \frac{2 \sin B \cos A}{2 \cos A \cos B} = \frac{\sin B}{\cos B} = \tan B$$

D. Identities in Numerical Problems

1.74: Identities in Numerical Problems

Identities can be applied to numerical problems.

Recognizing that an identity is to be applied can be the most difficult part.

Example 1.75

Given that $\tan \theta = \frac{\sqrt{40}}{3}$ evaluate:

$$\cos\left(\theta + \frac{\pi}{12}\right) \cos\left(\theta - \frac{\pi}{12}\right) - \sin\left(\theta + \frac{\pi}{12}\right) \sin\left(\theta - \frac{\pi}{12}\right)$$

Use a change of variable. Let $A = \theta + \frac{\pi}{12}$, $B = \theta - \frac{\pi}{12}$:

$$= \cos A \cos B - \sin A \sin B$$

Use the difference formula for $\cos$, and changing back to the original variable:

$$= \cos(A+B) = \cos\left(\theta + \frac{\pi}{12} + \theta - \frac{\pi}{12}\right) = \cos 2\theta = 2 \cos^2 \theta - 1$$

Substitute from a reference triangle $\tan \theta = \frac{\sqrt{40}}{3} \Rightarrow \cos \theta = \frac{3}{7} \Rightarrow \cos^2 \theta = \frac{9}{49}$

$$= 2\left(\frac{9}{49}\right) - 1 = \frac{18}{49} - 1 = -\frac{27}{49}$$

Example 1.76

$$Z = \frac{1}{2\sqrt{3}} \tan[\cos^{-1}(3X - 2Y)]$$

Find the solution set in terms of θ for the equation above given that:

$$Z = \sin\left(\frac{\pi}{12} - \theta\right) \cos\left(\frac{\pi}{12} + \theta\right) + \cos\left(\frac{\pi}{12} - \theta\right) \sin\left(\frac{\pi}{12} + \theta\right), \quad \theta \in \mathbb{R}$$

 $X = \text{Maximum value of } Z$ $Y = \text{Minimum value of } Z$ Use a change of variable. Let $A = \frac{\pi}{12} - \theta$, $B = \frac{\pi}{12} + \theta$

$$Z = \sin A \cos B + \cos A \sin B = \sin(A+B) = \sin\left(\frac{\pi}{12} - \theta + \frac{\pi}{12} + \theta\right) = \sin \frac{\pi}{6} = \frac{1}{2} = \text{Constant}$$

Since Z is a constant:

$$X = Y = \frac{1}{2} \Rightarrow 3X - 2Y = 3X - 2X = X = \frac{1}{2}$$

We can then show the given equation is an identity:

$$RHS = \frac{1}{2\sqrt{3}} \tan \left[\cos^{-1} \left(\frac{1}{2} \right) \right] = \frac{1}{2\sqrt{3}} \tan \left[\frac{\pi}{3} \right] = \frac{1}{2\sqrt{3}} \cdot \sqrt{3} = \frac{1}{2} = Z = LHS$$

Hence:

$$\theta \in \mathbb{R}$$

Example 1.77

$$2 \sin(\alpha - \theta) \cos \alpha = \sin(n\alpha - \theta) - \sin \theta$$

Given that the above is an identity determine the value of n , where n is a real number.

Use the difference identity on the LHS for $\sin(\alpha - \theta)$:

$$LHS = [2 \sin \alpha \cos \theta - 2 \cos \alpha \sin \theta] \cos \alpha$$

Distribute the $\cos \alpha$ to each term:

$$= 2 \sin \alpha \cos \alpha \cos \theta - 2 \cos^2 \alpha \sin \theta$$

Use the double angle identity to substitute $2 \sin \alpha \cos \alpha = 2 \sin 2\alpha$, $2 \cos^2 \alpha = \cos 2\alpha + 1$:

$$\begin{aligned} &= \sin 2\alpha \cos \theta - (\cos 2\alpha + 1) \sin \theta \\ &= \sin 2\alpha \cos \theta - \cos 2\alpha \sin \theta - \sin \theta \end{aligned}$$

Use the difference identity on the LHS:

$$= \sin(2\alpha - \theta) - \sin(\theta)$$

$$n = 2$$

1.5 More Identities

A. Sum of Sinusoidal Functions

1.78: Sum of Sinusoidal Functions

For all x and constants a, b, C it is true that:

$$a \sin x + b \cos x = A \sin(x + C)$$

Where:

$$\begin{aligned} A &= \sqrt{a^2 + b^2} \\ C &= \cos^{-1} \left(\frac{a}{\sqrt{a^2 + b^2}} \right) = \sin^{-1} \left(\frac{b}{\sqrt{a^2 + b^2}} \right) = \tan^{-1} \left(\frac{b}{a} \right) \end{aligned}$$

Suppose that for some constants A, B, C to be determined, it is true that:

$$a \sin x + b \cos x = A \sin(x + C)$$

Use the sum formula for $\sin$ on the RHS:

$$a \sin x + b \cos x = A \sin x \cos C + A \sin C \cos x$$

Use the method of undetermined coefficients. Comparing like terms:

$$A \cos C = a, \quad A \sin C = b$$

$$a^2 + b^2 = (A \cos c)^2 + (A \sin c)^2 = A^2 \cos^2 c + A^2 \sin^2 c = A^2 \Rightarrow A = \sqrt{a^2 + b^2}$$

Note that we did not take the negative square root because we did not need to.

$$\begin{aligned} \cos C &= \frac{a}{A} = \frac{a}{\sqrt{a^2 + b^2}} \Rightarrow C = \cos^{-1} \left(\frac{a}{\sqrt{a^2 + b^2}} \right) \\ \sin C &= \frac{b}{A} = \frac{b}{\sqrt{a^2 + b^2}} \Rightarrow C = \sin^{-1} \left(\frac{b}{\sqrt{a^2 + b^2}} \right) \\ \tan C &= \frac{\sin C}{\cos C} = \frac{\frac{b}{A}}{\frac{a}{A}} = \frac{b}{a} \Rightarrow C = \tan^{-1} \left(\frac{b}{a} \right) \end{aligned}$$

Example 1.79

Convert using the formula into a single *sin* function:

- A. $\sin x + \cos x$
- B. $\sin x - \sqrt{3} \cos x$

$$a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin \left(x + \tan^{-1} \left(\frac{b}{a} \right) \right)$$

Part A: Substitute $a = b = 1$:

$$\sin x + \cos x = \sqrt{1^2 + 1^2} \sin \left(x + \tan^{-1} \left(\frac{1}{1} \right) \right) = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right)$$

Part B: Substitute $a = 1, b = -\sqrt{3}$:

$$\sin x - \sqrt{3} \cos x = \sqrt{1^2 + (-\sqrt{3})^2} \sin \left(x + \tan^{-1} \left(-\frac{\sqrt{3}}{1} \right) \right) = 4 \sin \left(x - \frac{\pi}{3} \right)$$

Example 1.80

$$3 \sin x + 4 \cos x$$

Determine the

- A. Minimum value and maximum value
- B. Range
- C. Period

$$3 \sin x + 4 \cos x = \sqrt{3^2 + 4^2} \sin \left(x + \tan^{-1} \left(\frac{4}{3} \right) \right) = 5 \sin \left(x + \tan^{-1} \left(\frac{4}{3} \right) \right)$$

$$\text{Min Value} = -5$$

$$\text{Max Value} = 5$$

Range as two interpretations:

$$\text{Range of the function: } -5 \leq x \leq 5$$

$$\text{Range from statistics} = \text{Max} - \text{min} = 5 - (-5) = 10$$

Period

$$= \frac{2\pi}{1} = 2\pi$$

Example 1.81

Determine $\sin C$

$$2 \sin x + 3 \cos x = A \sin(Bx + C), C = \cos^{-1}\left(\frac{p}{\sqrt{q}}\right)$$

where the prime factorization of q has no perfect squares.

$$a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin\left(x + \cos^{-1}\left(\frac{a}{\sqrt{a^2 + b^2}}\right)\right)$$

Substitute $\sqrt{a^2 + b^2} = \sqrt{2^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13}$:

$$2 \sin x + 3 \cos x = \sqrt{13} \sin\left(x + \cos^{-1}\left(\frac{2}{\sqrt{13}}\right)\right)$$

$$C = \cos^{-1}\left(\frac{2}{\sqrt{13}}\right) \Rightarrow \cos C = \frac{2}{\sqrt{13}} \Rightarrow \sin C = \frac{3}{\sqrt{13}}$$

1.82: Sum of Sinusoidal Functions

For all x and constants a, b, C it is true that:

$$a \sin x + b \cos x = A \cos(x - C)$$

Where:

$$A = \sqrt{a^2 + b^2}$$

$$C = \cos^{-1}\left(\frac{a}{\sqrt{a^2 + b^2}}\right) = \sin^{-1}\left(\frac{b}{\sqrt{a^2 + b^2}}\right) = \tan^{-1}\left(\frac{b}{a}\right)$$

Suppose that for some constants A, B, C to be determined, it is true that:

$$a \sin x + b \cos x = A \cos(x - C)$$

Use the sum formula for $\cos$ on the RHS:

$$a \sin x + b \cos x = A \cos x \cos C + A \sin C \sin x$$

Use the method of undetermined coefficients. Comparing like terms:

$$A \cos C = b, \quad A \sin C = a$$

$$a^2 + b^2 = (A \sin C)^2 + (A \cos C)^2 = A^2 \sin^2 C + A^2 \cos^2 C = A^2 \Rightarrow A = \sqrt{a^2 + b^2}$$

Note that we did not take the negative square root because we did not need to.

$$\cos C = \frac{b}{A} = \frac{a}{\sqrt{a^2 + b^2}} \Rightarrow C = \cos^{-1}\left(\frac{b}{\sqrt{a^2 + b^2}}\right)$$

$$\sin C = \frac{a}{A} = \frac{b}{\sqrt{a^2 + b^2}} \Rightarrow C = \sin^{-1}\left(\frac{a}{\sqrt{a^2 + b^2}}\right)$$

$$\tan C = \frac{\sin C}{\cos C} = \frac{\frac{a}{\sqrt{a^2 + b^2}}}{\frac{b}{\sqrt{a^2 + b^2}}} = \frac{a}{b} \Rightarrow C = \tan^{-1}\left(\frac{a}{b}\right)$$

B. Conditional Identities

A conditional identity is an identity that is true under specific conditions.

1.83: Angles of a Triangle

If three angles A , B and C are the angles of a triangle, then their sum is:

$$A + B + C = \pi$$

The sum of the angle of a triangle is $180^\circ = \pi$.

$$A + B + C = 180^\circ = \pi$$

Hence, the result

Example 1.84

Given $\triangle ABC$ with angles A , B and C show that:

$$\sin(A + B) = \sin C$$

Method I: Begin from LHS

$$LHS = \sin(A + B) = \sin(A + B + C - C) = \sin(\pi - C) = \sin C = RHS$$

Method I: Begin from RHS

$$\begin{aligned} A + B + C = \pi &\Rightarrow C = \pi - (A + B) \\ RHS = \sin C &= \sin[\pi - (A + B)] = \sin(A + B) = LHS \end{aligned}$$

Example 1.85

Given $\triangle ABC$ with angles A , B and C show that:

$$\cos C = \sin A \sin B - \cos A \cos B$$

$$\begin{aligned} RHS &= -(\cos A \cos B - \sin A \sin B) \\ &= -\cos(A + B) \\ &= -\cos(A + B + C - C) \\ &= -\cos(\pi - C) \\ &= \cos C = LHS \end{aligned}$$

Example 1.86

Given $\triangle ABC$ with angles A , B and C show that:

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

Method I: Clever Method

We have been given that:

$$A + B + C = \pi \Rightarrow A + B = \pi - C$$

Take the tan of both sides, and apply the sum formula for tan:

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(A + B) = \tan(\pi - C) = -\tan C$$

Equating the first expression with the last:

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

Cross-multiply:

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

Which is the result we wanted.

Method II: Brute Force

$$LHS = \tan A + \tan B + \tan C$$

$$\begin{aligned} \text{Substitute } \tan C &= \tan(\pi - (A + B)) = -\tan A + B \\ &= \tan A + \tan B - \tan(A + B) \\ &= \tan A + \tan B - \left[\frac{\tan A + \tan B}{1 - \tan A \tan B} \right] \\ &= \frac{1(\tan A + \tan B) - \tan A \tan B (\tan A + \tan B) - (\tan A + \tan B)}{\tan A \tan B - 1} \\ &= -\tan A \tan B \left[\frac{\tan A + \tan B}{1 - \tan A \tan B} \right] \end{aligned}$$

$$\begin{aligned} \text{Substitute } \frac{\tan A + \tan B}{1 - \tan A \tan B} &= \tan(A + B) = -\tan C: \\ &= -\tan A \tan B [-\tan C] \\ &= \tan A \tan B \tan C \end{aligned}$$

Example 1.87

Given $\triangle ABC$ with angles A, B and C show that:

$$\cos A \cos B \cos C - \sin A \sin B \cos C = \sin A \cos B \sin C + \cos A \sin B \sin C - 1$$

Factor $\cos C$ in the LHS:

$$\begin{aligned} LHS &= \cos C (\cos A \cos B - \sin A \sin B) \\ &= \cos C (\cos A \cos B - \sin A \sin B) \\ &= \cos C \cos(A + B) \\ &= \cos C \cos(\pi - C) \\ &= -\cos^2 C \end{aligned}$$

Factor $\sin C$ in the RHS:

$$\begin{aligned} \sin C (\sin A \cos B + \cos A \sin B) - 1 \\ &= \sin C \sin(A + B) - 1 \\ &= \sin C \sin(\pi - C) - 1 \\ &= \sin^2 C - 1 \\ &= -\cos^2 C = LHS \end{aligned}$$

Example 1.88

$$\tan\left(x + \frac{\pi}{2}\right) = -\cot x$$

Expand the LHS

$$\begin{aligned} LHS &= \frac{\sin\left(x + \frac{\pi}{2}\right)}{\cos\left(x + \frac{\pi}{2}\right)} = \frac{\sin x \cos\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right) \cos x}{\cos x \cos\left(\frac{\pi}{2}\right) - \sin x \sin\left(\frac{\pi}{2}\right)} \\ &= \frac{\sin x (0) + (1) \cos x}{\cos x (0) - \sin x (1)} = \frac{\cos x}{-\sin x} = -\cot x \end{aligned}$$

C. Reduction Identities**1.89: Triple Angle Identity**

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

Split $3x$ as $2x + x$ in the LHS:

$$\sin 3x = \sin(2x + x)$$

Use the sum identity:

$$= \sin 2x \cos x + \sin x \cos 2x$$

Use the double angle identities:

$$\begin{aligned} &= (2 \sin x \cos x) \cos x + \sin x (1 - 2 \sin^2 x) \\ &= 2 \sin x \cos^2 x + \sin x - 2 \sin^3 x \end{aligned}$$

Use the Pythagorean Identity:

$$\begin{aligned} &= 2 \sin x (1 - \sin^2 x) + \sin x - 2 \sin^3 x \\ &= 2 \sin x - 2 \sin^3 x + \sin x - 2 \sin^3 x \\ &= 3 \sin x - 4 \sin^3 x = RHS \end{aligned}$$

D. Product to Sum Identities

1.90: Product to Sum Formula

$$\begin{aligned} 2 \sin \alpha \cos \beta &= \sin(\alpha + \beta) + \sin(\alpha - \beta) \\ 2 \cos \alpha \sin \beta &= \sin(\alpha + \beta) - \sin(\alpha - \beta) = \end{aligned}$$

We can set up some cancellation here by expanding using the formula for the sum and differences of sines:

$$\begin{aligned} &\underbrace{\sin \alpha \cos \beta + \cos \alpha \sin \beta}_{\sin(\alpha+\beta)} + \underbrace{\sin \alpha \cos \beta - \cos \alpha \sin \beta}_{\sin(\alpha-\beta)} = 2 \sin \alpha \cos \beta \\ &\underbrace{\sin \alpha \cos \beta + \cos \alpha \sin \beta}_{\sin(\alpha+\beta)} - \underbrace{\sin \alpha \cos \beta - \cos \alpha \sin \beta}_{\sin(\alpha-\beta)} = 2 \cos \alpha \sin \beta \end{aligned}$$

Example 1.91

Product to Sum Formula: $2 \sin \alpha \cos \beta = \sin(\alpha + \beta) + \sin(\alpha - \beta)$

Prove the product to sum formula using the sum to product formula.

Apply the sum to product formula on the RHS:

$$\begin{aligned} RHS &= 2 \left[\sin \left(\frac{(\alpha + \beta) + (\alpha - \beta)}{2} \right) \cos \left(\frac{(\alpha + \beta) - (\alpha - \beta)}{2} \right) \right] \\ &= 2 \left[\sin \left(\frac{2\alpha}{2} \right) \cos \left(\frac{2\beta}{2} \right) \right] \\ &= 2 [\sin \alpha \cos \beta] \end{aligned}$$

E. Sum to Product Identities

1.92: $\cos$

$$\cos \alpha + \cos \beta = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$$

F. A product identity

1.93: Product

$$\sin\left(\frac{\pi}{n}\right) \sin\left(\frac{2\pi}{n}\right) \sin\left(\frac{3\pi}{n}\right) \dots \sin\left(\frac{(n-1)\pi}{n}\right) = \frac{n}{2^{n-1}}, \quad n \in \mathbb{N}$$

Example 1.94

$$\sin\left(\frac{\pi}{8}\right) \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{3\pi}{8}\right) \sin\left(\frac{\pi}{2}\right) \dots \sin\left(\frac{7\pi}{8}\right)$$

$$\sin\left(\frac{\pi}{8}\right) \sin\left(\frac{2\pi}{8}\right) \sin\left(\frac{3\pi}{8}\right) \sin\left(\frac{4\pi}{8}\right) \dots \sin\left(\frac{7\pi}{8}\right)$$

Substitute $n = 8 = 2^3$ in $\frac{n}{2^{n-1}}$:

$$\frac{2^3}{2^{8-1}} = \frac{2^3}{2^7} = \frac{1}{2^4} = \frac{1}{16}$$

Example 1.95

$$a = \sin\left(\frac{\pi}{16}\right) \sin\left(\frac{2\pi}{16}\right) \sin\left(\frac{3\pi}{16}\right) \dots \sin\left(\frac{15\pi}{16}\right)$$

$$b = \sin\left(\frac{17\pi}{16}\right) \sin\left(\frac{18\pi}{16}\right) \dots \sin\left(\frac{31\pi}{16}\right)$$

Find ab

Substitute $n = 16 = 2^4$ in $\frac{n}{2^{n-1}}$:

$$a = \sin\left(\frac{\pi}{16}\right) \sin\left(\frac{2\pi}{16}\right) \sin\left(\frac{3\pi}{16}\right) \dots \sin\left(\frac{15\pi}{16}\right) = \frac{2^4}{2^{16-1}} = \frac{2^4}{2^{15}} = \frac{1}{2^{11}}$$

Substitute $n = 16 = 2^4$ in $\frac{n}{2^{n-1}}$:

$$\begin{aligned} b &= \sin\left(\pi + \frac{\pi}{16}\right) \sin\left(\pi + \frac{2\pi}{16}\right) \dots \sin\left(\pi + \frac{15\pi}{16}\right) \\ &= \left[-\sin\left(\frac{\pi}{16}\right)\right] \left[-\sin\left(\frac{2\pi}{16}\right)\right] \dots \left[-\sin\left(\frac{15\pi}{16}\right)\right] \\ &= -\sin\left(\frac{\pi}{16}\right) \sin\left(\frac{2\pi}{16}\right) \sin\left(\frac{3\pi}{16}\right) \dots \sin\left(\frac{15\pi}{16}\right) \\ &= -a \end{aligned}$$

$$ab = -a^2 = -\left(\frac{1}{2^{11}}\right)^2 = -\frac{1}{2^{22}}$$

1.96: Product Notation

$$\prod_{k=1}^{k=n-1} \sin\left(\frac{k\pi}{n}\right) = \frac{n}{2^{n-1}}, \quad n \in \mathbb{N}$$

Example 1.97

$$\sin\left(\frac{\pi}{7}\right) \cdot \sin\left(\frac{2\pi}{7}\right) \cdot \dots \cdot \sin\left(\frac{6\pi}{7}\right)$$

The property above is applicable with $n = 7$:

$$\prod_{k=1}^{k=6} \sin\left(\frac{k\pi}{7}\right) = \frac{7}{2^6} = \frac{7}{64}$$

Example 1.98

Determine the value of the expression given that it has n terms:

$$\sin\left(\frac{\pi}{n}\right) \cdot \sin\left(\frac{2\pi}{n}\right) \cdot \sin\left(\frac{3\pi}{n}\right) \cdot \dots \cdot$$

$$\sin\left(\frac{\pi}{n}\right) \cdot \sin\left(\frac{2\pi}{n}\right) \cdot \sin\left(\frac{3\pi}{n}\right) \cdot \dots \cdot \sin\left(\frac{n\pi}{n}\right)$$

$$\sin\left(\frac{n\pi}{n}\right) = \sin \pi = 0$$

$$\text{Expression} = 0$$

Example 1.99³

Determine the value of the expression below as a real number:

$$\sin\left(\frac{\pi}{7}\right) \cdot \sin\left(\frac{2\pi}{7}\right) \cdot \sin\left(\frac{3\pi}{7}\right)$$

Let

$$x = \sin\left(\frac{\pi}{7}\right) \cdot \sin\left(\frac{2\pi}{7}\right) \cdot \sin\left(\frac{3\pi}{7}\right)$$

Square both sides:

$$x^2 = \sin\left(\frac{\pi}{7}\right) \cdot \sin\left(\frac{2\pi}{7}\right) \cdot \sin\left(\frac{3\pi}{7}\right) \cdot \sin\left(\frac{\pi}{7}\right) \cdot \sin\left(\frac{2\pi}{7}\right) \cdot \sin\left(\frac{3\pi}{7}\right)$$

Use the property $\sin \theta = \sin(\pi - \theta)$ on the last three terms:

$$\sin\left(\frac{\pi}{7}\right) = \sin\left(\pi - \frac{\pi}{7}\right) = \sin\left(\frac{6\pi}{7}\right)$$

$$\sin\left(\frac{2\pi}{7}\right) = \sin\left(\pi - \frac{2\pi}{7}\right) = \sin\left(\frac{5\pi}{7}\right)$$

$$\sin\left(\frac{3\pi}{7}\right) = \sin\left(\pi - \frac{3\pi}{7}\right) = \sin\left(\frac{4\pi}{7}\right)$$

Make the above substitutions:

$$x^2 = \sin\left(\frac{\pi}{7}\right) \cdot \sin\left(\frac{2\pi}{7}\right) \cdot \sin\left(\frac{3\pi}{7}\right) \cdot \sin\left(\frac{4\pi}{7}\right) \cdot \sin\left(\frac{5\pi}{7}\right) \cdot \sin\left(\frac{6\pi}{7}\right)$$

$$x^2 = \frac{7}{64}$$

$$x = \frac{\sqrt{7}}{8}$$

³ This [lengthier explanation](#) makes use only of standard trigonometric properties.

100 Examples